If you're wondering, when Earth passes in front of the Sun—which is always happening for some line of sight in the galaxy—we block 1/10,000th of the Sun's surface, thereby briefly dimming the Sun's total light by 1/10,000th of its normal brightness. Fine as it goes. They'll discover that Earth exists, but learn nothing about happenings on Earth's surface.

Radio waves and microwaves might work. Maybe our eavesdropping aliens have something like the 500-meter radio telescope in the Guizhou province of China. If they do, and if they tune to the right frequencies, they'll certainly notice Earth—or rather, they'll notice our modern civilization as one of the most luminous sources in the sky. Consider everything we've got that generates radio waves and microwaves: not only traditional radio itself, but also broadcast television, mobile phones, microwave ovens, garage-door openers, car-door unlockers, commercial radar, military radar, and communications satellites. We're ablaze in long-frequency waves—spectacular evidence that something unusual is going on here, because in their natural state, small rocky planets emit hardly any radio waves at all.

So if those alien eavesdroppers turn their own version of a radio telescope in our direction, they might infer that our planet hosts technology. One complication, though: other interpretations are possible. Maybe they wouldn't be able to distinguish Earth's signals from those of the larger planets in our solar system, all of which are sizable sources of radio waves, especially Jupiter. Maybe they'd think we were a new kind of odd, radio-intensive planet. Maybe they wouldn't be able to distinguish Earth's radio emissions from those of the Sun, forcing them to conclude that the Sun is a new kind of odd, radio-intensive star.

Astrophysicists right here on Earth, at the University of Cambridge in England, were similarly stumped back in 1967. While surveying the skies with a radio telescope for any source of strong radio waves, Antony Hewish and his team discovered something extremely odd: an object pulsing at precise, repeating intervals of slightly more than a second. Jocelyn Bell, a graduate student of Hewish's at the time, was the first to notice it.

Soon Bell's colleagues established that the pulses came from a great distance. The thought that the signal was technological—another culture beaming evidence of its activities across space—was irresistible. As Bell recounts, "We had no proof that it was an entirely natural radio emission. . . . Here was I trying to get a Ph.D. out of a new technique, and some silly lot of little green men had to choose my aerial and my frequency to